COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling

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Multiple Linear Regression Model

Model with p predictors and an intercept:

$$\boxed{Y_i = \beta_0 + \beta_1 X_{i1} + \ \dots + \beta_p X_{ip} + \varepsilon_i, \qquad i = 1, \dots, n}$$

In matrix form: $\boldsymbol{Y}$ is $n\times 1$ output vector and $\boldsymbol{X}$ is $n\times(p+1)$ input data matrix

$$\boldsymbol{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \qquad \boldsymbol{X} = \begin{bmatrix} 1 & X_{11} & \dots & X_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \dots & X_{np} \end{bmatrix}$$

Then with β being $p\times 1$ vector and $\boldsymbol{\varepsilon}$ an $n\times 1$ vector, we have:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}, \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}, \mathrm{E}(\boldsymbol{\varepsilon}) = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \mathrm{cov}(\boldsymbol{\varepsilon}) = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix} = \sigma^2 \boldsymbol{I}_n$$

Model in matrix form:

$$\boxed{\begin{gathered} \boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon} \\ \Rightarrow \mathrm{E}(\boldsymbol{Y}|\boldsymbol{X}) = \boldsymbol{X\beta}, \qquad \mathrm{cov}(\boldsymbol{Y}|\boldsymbol{X}) = \sigma^2 \boldsymbol{I}_n \end{gathered}}$$

Least Squares Estimator

Define the *sum of square errors* as:

$$\boxed{SSE = Q(\boldsymbol{\beta}) = \sum_{i=1}^{n}\left(Y_i - \beta_0 - \beta_1 X_{i1} - \cdots \beta_p X_{ip}\right)^2 = (\boldsymbol{Y} - \boldsymbol{X\beta})'(\boldsymbol{Y} - \boldsymbol{X\beta})}$$

Goal: Obtain the solution $\boldsymbol{b}$ which minimizes $Q(\boldsymbol{\beta})$

It can be shown that

$$Q(\boldsymbol{\beta}) = \boldsymbol{Y}'\boldsymbol{Y} - 2\boldsymbol{Y}'\boldsymbol{X\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X\beta}$$

Aside: Multivariable calculus results (assume $\boldsymbol{A}$ is symmetric for second result)

$$\frac{\partial \boldsymbol{Ax}}{\partial \boldsymbol{x}} = \boldsymbol{A}$$

$$\frac{\partial \boldsymbol{x}'\boldsymbol{Ax}}{\partial \boldsymbol{x}} = 2\boldsymbol{x}'\boldsymbol{A}$$

$$\Rightarrow \frac{\partial Q(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\boldsymbol{Y}'\boldsymbol{X} + 2\boldsymbol{X}'\boldsymbol{X\beta} = \boldsymbol{0} \Leftrightarrow \boldsymbol{X}'\boldsymbol{X\beta} = \boldsymbol{Y}'\boldsymbol{X}$$

$$\boxed{\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{Y}'\boldsymbol{X}}$$

Fitted Values and Residuals

Fitted values $\widehat{\boldsymbol{Y}}$ and residuals $\boldsymbol{e}$ are $n\times 1$ vectors:

$$\boxed{\widehat{\boldsymbol{Y}} = \boldsymbol{X}\boldsymbol{b} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} = \boldsymbol{H}\boldsymbol{Y}}$$

where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is the so-called "hat matrix".

$$\boxed{\boldsymbol{e} = \boldsymbol{Y} - \widehat{\boldsymbol{Y}} = \boldsymbol{Y} - \boldsymbol{X}\boldsymbol{b} = \boldsymbol{Y} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y}}$$

Their covariance matrices are:

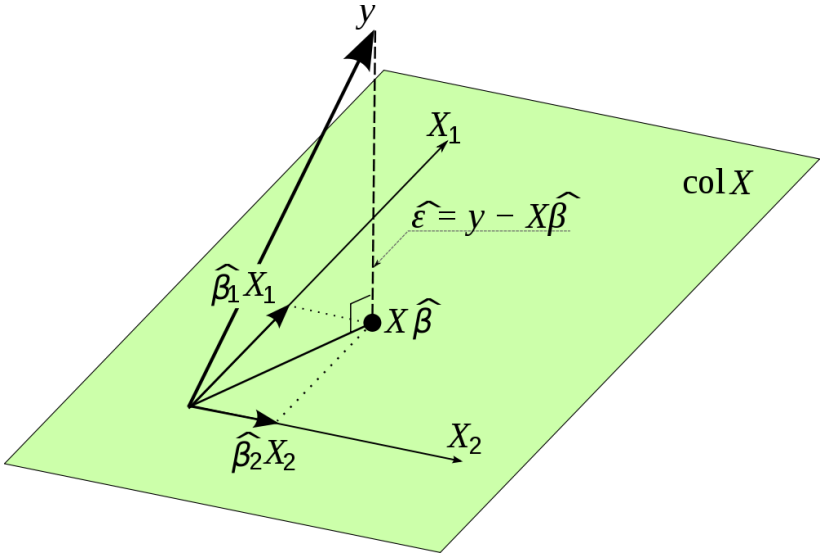
$$\mathrm{cov}\left(\widehat{\boldsymbol{Y}}\right) = \mathrm{cov}(\boldsymbol{H}\boldsymbol{Y}) = \boldsymbol{H}\mathrm{cov}(\boldsymbol{Y})\boldsymbol{H}' = \sigma^2\boldsymbol{H}$$

Similarly, for the residuals:

$$\mathrm{cov}(\boldsymbol{e}) = \mathrm{cov}((\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y}) = (\boldsymbol{I} - \boldsymbol{H})\mathrm{cov}(\boldsymbol{Y})(\boldsymbol{I} - \boldsymbol{H})' = \sigma^2(\boldsymbol{I} - \boldsymbol{H})$$

How does this work?

$$\widehat{\boldsymbol{Y}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y} = \boldsymbol{H}\boldsymbol{Y},$$
where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$



The matrix $\boldsymbol{H}$ is called the "hat" or "projection" matrix. Note that it is *idempotent* ($\boldsymbol{HH} = \boldsymbol{H}$) and symmetric. This also means $\boldsymbol{H}$ is orthogonal ($\boldsymbol{H}' = \boldsymbol{H}^{-1}$), so it can be written as $\boldsymbol{H} = \boldsymbol{UU}'$, where $\boldsymbol{U} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-\frac{1}{2}}$

This means $\boldsymbol{U}$ is a new orthonormal basis in $\mathbb{R}^{p+1}$, a linear subspace of $\mathbb{R}^n$ and $\boldsymbol{U}'\boldsymbol{Y}$ is the projection of $\boldsymbol{Y}$ on this new orthonormal basis.

If we don't stick to orthonormal basis, then the LS solutions $\boldsymbol{b}$ are the coordinates of the projection onto the subspace spanned by the columns of $\boldsymbol{X}$.

Inference

- $\boldsymbol{b}$ is an unbiased estimator of $\boldsymbol{\beta}$

Proof:

$$E(\boldsymbol{b}) = E[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathrm{E}(\boldsymbol{Y}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}$$

- The covariance matrix of $\boldsymbol{b}$ is:

$$\boxed{\mathrm{cov}(\boldsymbol{b}) = \sigma^2 \times (\boldsymbol{X}'\boldsymbol{X})^{-1}}$$

Proof:

$$\mathrm{cov}(\boldsymbol{b}) = \mathrm{cov}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}] = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathrm{cov}(\boldsymbol{Y})[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']'$$
$$= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Note: From the above we can estimate $SE(b_k) = \sqrt{Var(b_k)} = \sqrt{diag_k(\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1})}$

- To test significance of kth predictor H_0: $\beta_k = \beta_k^0$ vs. H_a: $\beta_k \neq \beta_k^0$, the test is

$$\frac{b_k - \beta_k^0}{SE(b_k)} \sim t_{n-p-1}$$

- $(100-\alpha)\%$ CI for β_k is $\boxed{b_k \pm t_{\frac{\alpha}{2},n-p-1} \times SE(b_k)}$

Distribution Results Details

Assume $\boldsymbol{\varepsilon}|\boldsymbol{X} \sim N_n(0, \sigma^2 \boldsymbol{I}_n)$ and we use p predictors. Then:

$$\boldsymbol{y}|\boldsymbol{X} \sim N_n(\boldsymbol{X\beta}, \sigma^2 \boldsymbol{I}_n)$$

$$\boxed{\boldsymbol{b} \sim N_{p+1}(\boldsymbol{\beta}, \sigma^2 (\boldsymbol{X}'\boldsymbol{X})^{-1})}$$

$\hat{\sigma}^2 = \frac{SSE}{n-p-1}$ is independent of $\boldsymbol{b}$, where $SSE = \boldsymbol{y}'[\boldsymbol{I} - \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}']\boldsymbol{y}$

$$\boxed{\frac{SSE}{\sigma^2} \sim \chi^2_{n-p-1}}$$

Proof: The above follows from (see p. 57 of Searle).

Theorem: When $\boldsymbol{y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\boldsymbol{y}'\boldsymbol{A}\boldsymbol{y} \sim \chi^2_{rank(A)}\left(\frac{1}{2}\boldsymbol{\mu}'\boldsymbol{A}\boldsymbol{\mu}\right)$ iff $\boldsymbol{A\Sigma}$ is idempotent, where $\frac{1}{2}\boldsymbol{\mu}'\boldsymbol{A}\boldsymbol{\mu}$ is non-centrality parameter.

ANOVA Approach to Regression

Let n be the sample size and p the number of predictors.
Analysis of Variance Table:

Source	SS	df	MS	F
Regression	$\text{SSR} = \sum(\hat{Y}_i - \bar{Y})^2$	p	$\text{MSR} = \frac{\text{SSR}}{p}$	$F = \frac{\text{MSR}}{\text{MSE}}$
Error	$\text{SSE} = \sum(Y_i - \hat{Y}_i)^2$	$n - p - 1$	$\text{MSE} = \frac{\text{SSE}}{n - p - 1}$	
Total	$\text{SST} = \sum(Y_i - \bar{Y})^2$	$n - 1$		

All the different sum of squares are quadratic forms in $\boldsymbol{Y}$ and

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}}$$

F-test and R^2

To test the *overall* significance of the model:

$$H_0: \beta_1 = \ldots = \beta_p = 0 \text{ vs. } H_a: \text{Not all } \beta_j = 0$$

The test statistic is

$$F = \frac{MSR}{MSE} \sim F_{p,n-p-1}$$

Measure of goodness of fit:

$$\boxed{R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}}$$

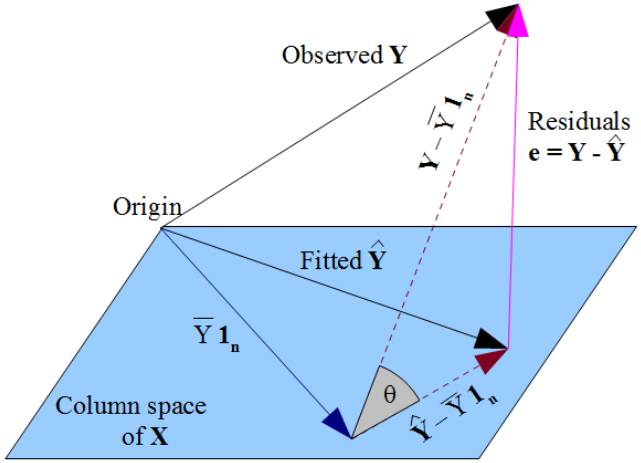
Side result:

$$F = \frac{R^2}{1 - R^2} \times \frac{n - p - 1}{p}$$

Adjusted for the number of predictors:

$$\boxed{R^2_{adj} = 1 - \left(\frac{n-1}{n-p}\right)\frac{SSE}{SST}}$$

Back to the Projections



The decomposition

$$\text{SST} = \text{SSR} + \text{SSE}$$

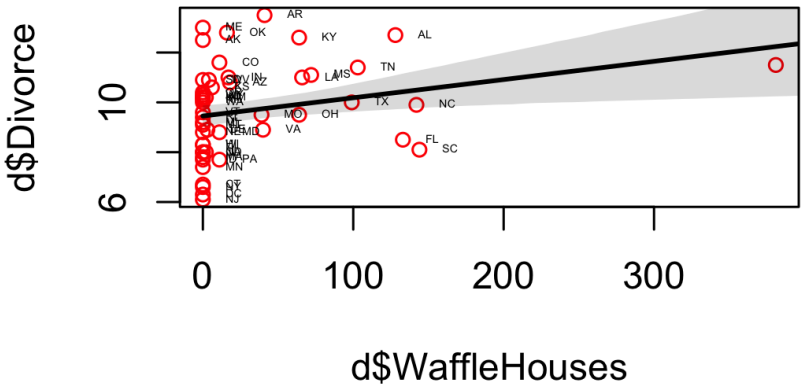
is just the Pythagorean theorem: $\|\boldsymbol{Y} - \bar{Y}\mathbf{1}\| = \left\|\widehat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}\right\|^2 + \left\|\boldsymbol{Y} - \hat{Y}\right\|^2$

$$\sqrt{R^2} = \sqrt{\frac{SSR}{SST}} = \cos(\theta) = \frac{\left\|\widehat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}\right\|}{\|\boldsymbol{Y} - \bar{Y}\mathbf{1}\|}$$

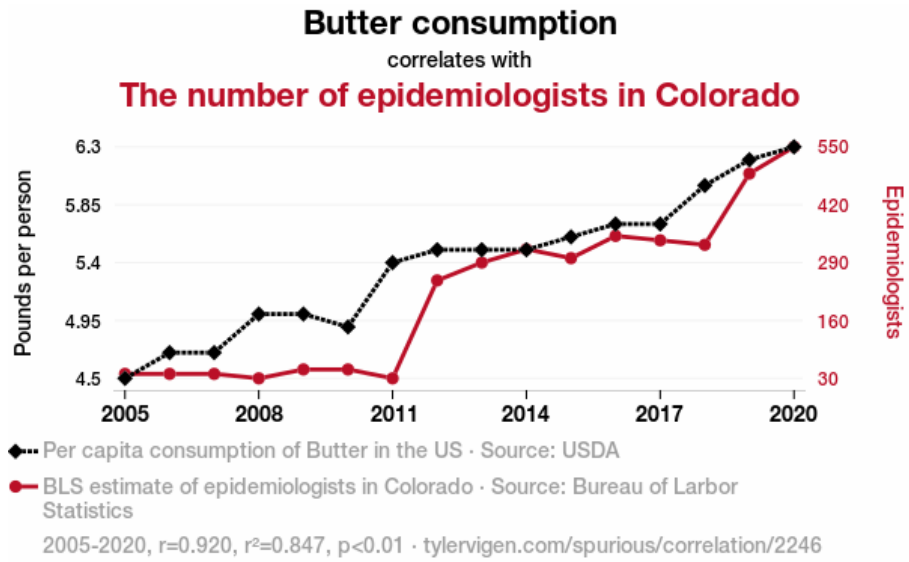
Therefore

$$R^2 = 1 - \frac{\left\|Y - \hat{Y}\right\|^2}{\|Y - \bar{Y}\mathbf{1}\|^2} = 1 - \sin^2\theta = \cos^2\theta = \frac{\left\|\hat{Y} - \bar{Y}\mathbf{1}\right\|^2}{\|Y - \bar{Y}\mathbf{1}\|^2}$$

Example:
Does Waffle House cause?



Correlation is commonplace



- http://www.tylervigen.com/spurious-correlations
- AI_explanation:

As butter consumption in Colorado increased, so did the demand for larger sample sizes in epidemiological studies. This led to a surge in funding for public health research, prompting a migration of epidemiologists to the state, lured by the opportunity to conduct margarineally significant studies and spread their knowledge like, well, butter.

Show prompt used to generate this paper

The Butter Effect: A Statistical Analysis of Butter Consumption and the Proportion of Epidemiologists in Colorado

The Journal of Irreverent Epidemiology

Jan 2024

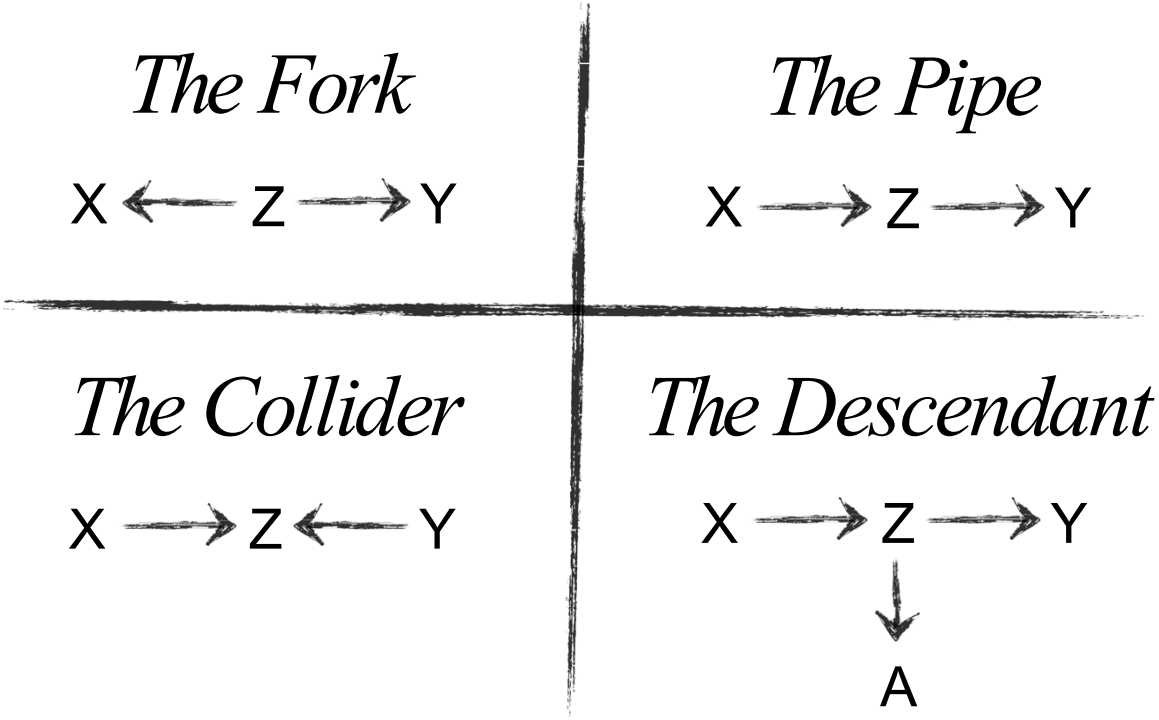
Reminder: This paper is AI-generated. Not real!

How this works

Data dredging: They have 25,153 variables in a database. They compare all these variables against each other to find ones that randomly match up. That's 632,673,409 correlation calculations! This is called "data dredging."

Common Elemental Confounds

Sometimes factors other than the IV may influence the DV in an experiment. These unwanted influences are called confounding variables.



The Fork

$$X \leftarrow Z \rightarrow Y$$

X and Y are associated

Z is a "confounder"

Share a common cause Z

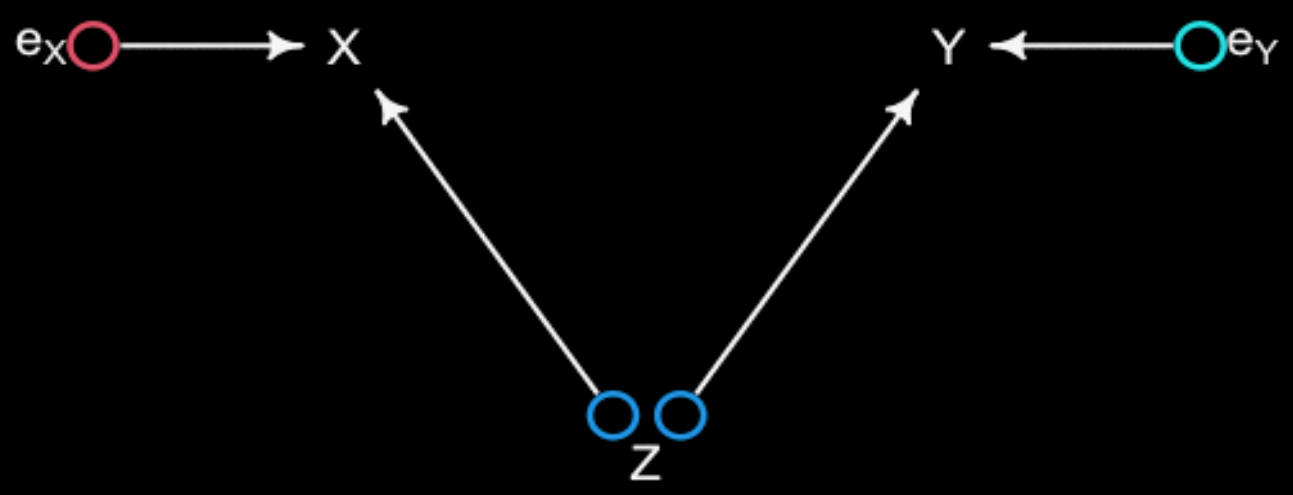
Once stratified by Z, no association

$$Y \not\perp\!\!\!\perp X$$

$$Y \perp\!\!\!\perp X \mid Z$$

Example:
Chocolate consumption $\leftarrow$ GDP $\rightarrow$ Nobel Prize count
If GDP stays constant, eating more chocolate would not increase the expected number of Nobel Prizes.

How to simulate the Fork



$$X \leftarrow Z \rightarrow Y$$

```
N <- 400
Z <- rbern(N)
X <- rnorm(N,2*Z-1)
Y <- rnorm(N,2*Z-1)
```

```
      Y
X      0   1
  0  397  84
  1  100 419
```

$$Y \not\perp\!\!\!\perp X$$

```
> cor(X,Y)
[1] 0.63
```

```
Z = 0
      Y
X      0   1
  0  390  43
  1   44   5
```

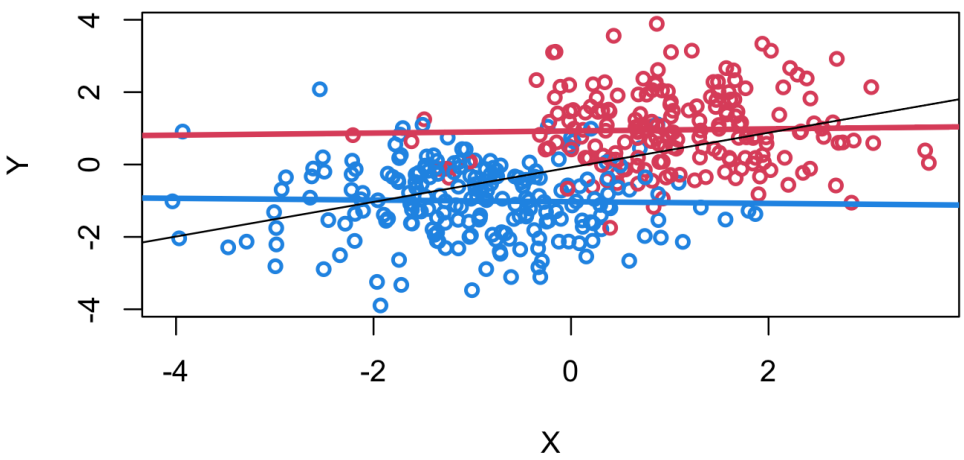
$$Y \perp\!\!\!\perp X \mid Z$$

```
Z = 1
      Y
X      0   1
  0    7  41
  1   56 414
```

```
> cor(X[Z==0],Y[Z==0])
[1] 0.003
> cor(X[Z==1],Y[Z==1])
[1] 0.024
```

Simulating a fork $X \leftarrow Z \rightarrow Y$

```
cols = c(4,2)

N <- 300
Z <- rbern(N)
X <- rnorm(N,2*Z-1) Y <-
rnorm(N,2*Z-1)

plot( X , Y , col=cols[Z+1] , lwd=3 )

abline(lm(Y[Z==1]~X[Z==1]),col=2,lwd=3)

abline(lm(Y[Z==0]~X[Z==0]),col=4,lwd=3)

abline(lm(Y~X),lwd=3)
```



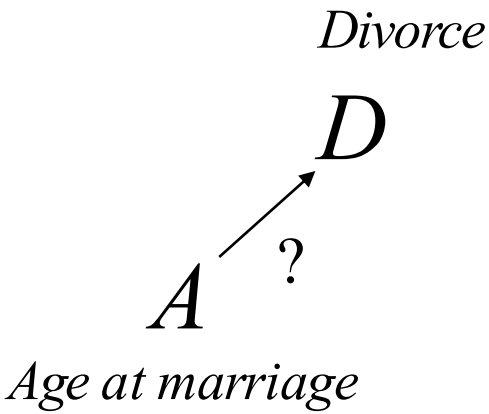

```
> t.test(Y[Z==1], Y[Z==0])

        Welch Two Sample t-test

data:  Y[Z == 1] and Y[Z == 0]
t = 19.314, df = 388.36, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.764692 2.164689
sample estimates:
 mean of x  mean of y
 0.9596821 -1.0050083
```

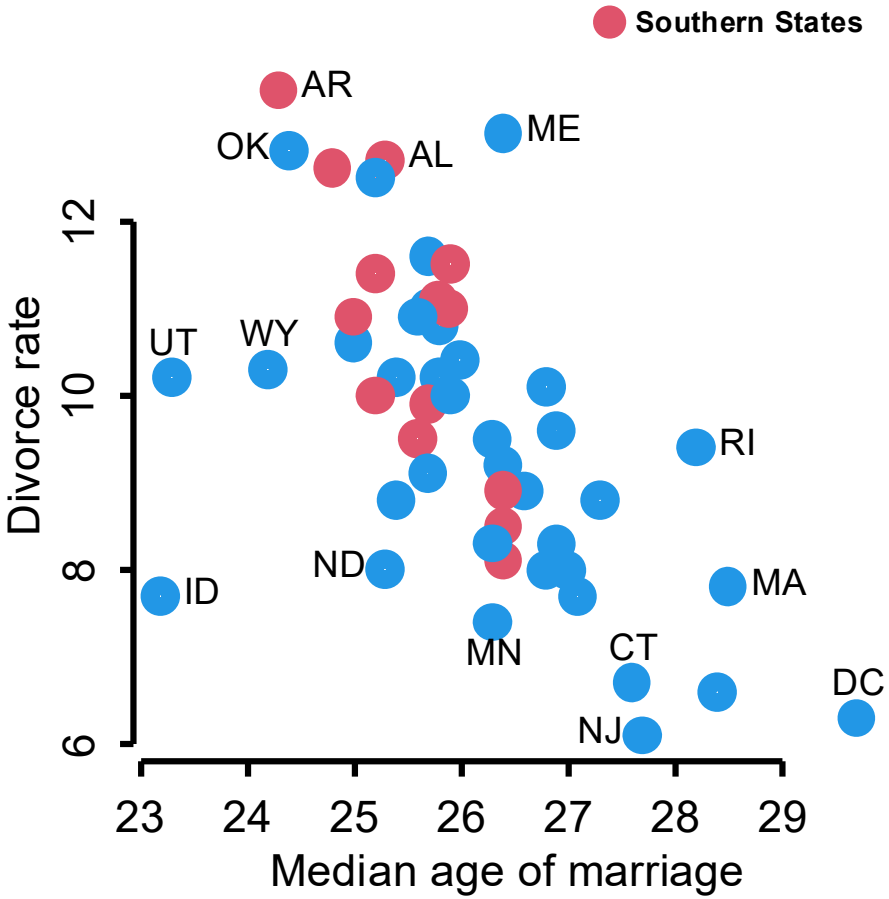
Fork Example

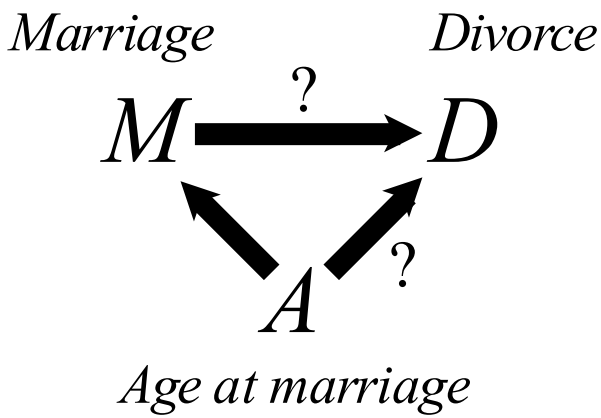
```
library(rethinking)
data(WaffleDivorce)
```

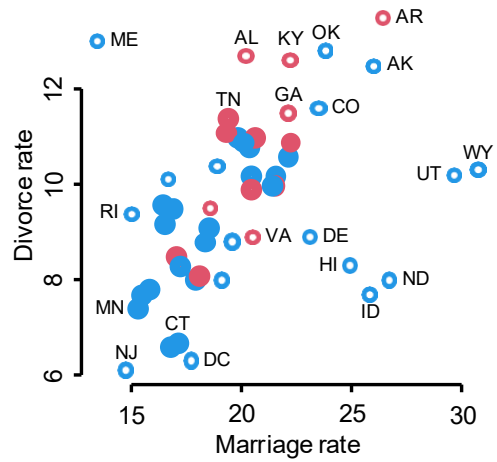
Why do regions of the USA with higher rates of **marriage** also have higher rates of **divorce**?

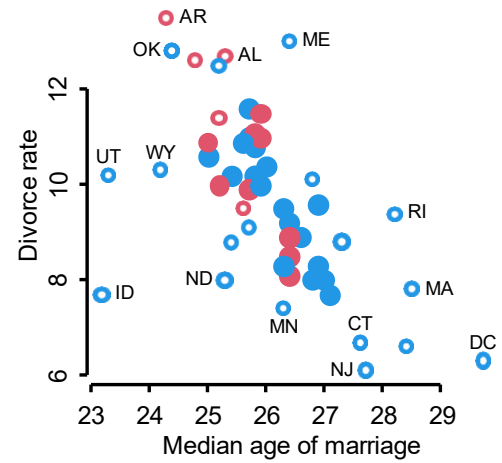
$$M \xrightarrow{?} D$$


Divorce
D
?
A
Age at marriage


Southern States
AR
ME
OK
AL
UT
WY
RI
ND
ID
MA
MN
CT
DC
NJ
Divorce rate
6
8
10
12
23
24
25
26
27
28
29
Median age of marriage



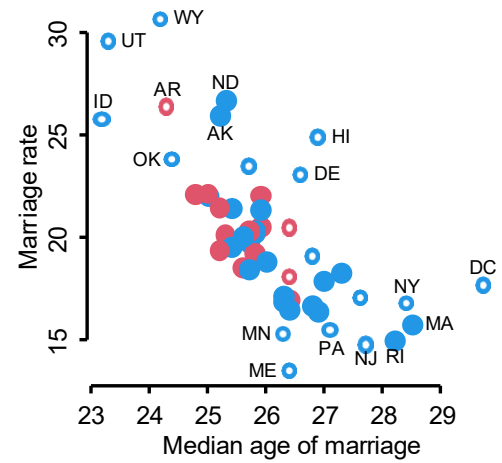




Fork: ***M*** <– ***A*** –> ***D***

To estimate **direct effect** of ***M***, need to break the fork

Break the fork by **stratifying** by ***A***

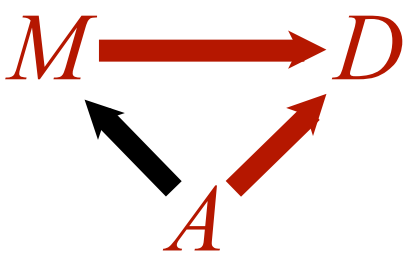


What does it mean to stratify by a continuous variable?

Categorical predictor: easy (case by case)

How does A influence D? What is $D = f(A,M)$?

In a linear regression:



$$D_i \sim \text{Normal}(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta_M M_i + \beta_A A_i$$

Interpretations:

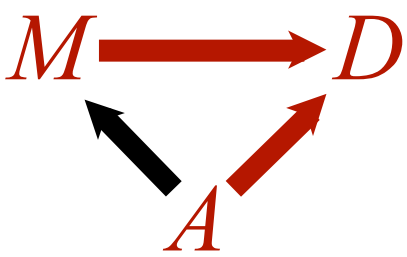
β_A is the effect of A, <u>*keeping M fixed*</u>!

β_M is the effect of M, <u>*keeping A fixed*</u>!

What does it mean to stratify by a continuous variable?

$$\mu_i = \alpha + \beta_M M_i + \beta_A A_i$$

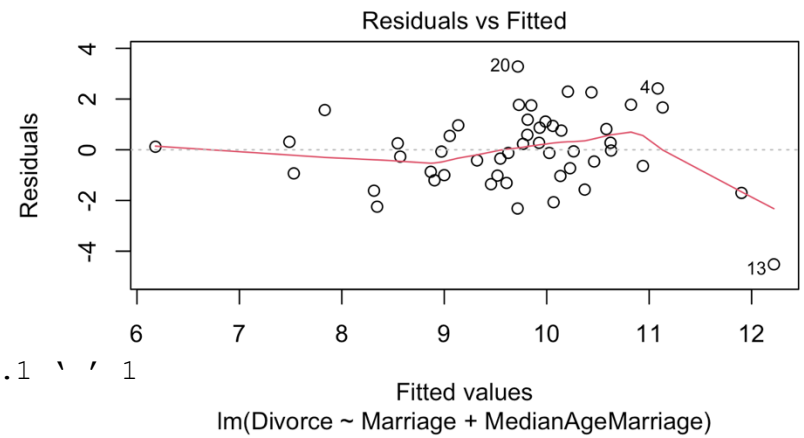
Every value of A produces a different relationship between D and M:



$$\mu_i = \underbrace{(\alpha + \beta_A A_i)}_{intercept} + \beta_M M_i$$

Analyze data

```
> reg2 = lm(Divorce ~ Marriage + MedianAgeMarriage, data = d)
> summary(reg2)
Call:
lm(formula = Divorce ~ Marriage + MedianAgeMarriage, data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5177 -0.9828 -0.0458  0.9224  3.2818
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       36.87665    7.66104   4.814 1.58e-05
Marriage          -0.05686    0.08053  -0.706 0.483594
MedianAgeMarriage -0.99965    0.24593  -4.065 0.000182
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.483 on 47 degrees of freedom
Multiple R-squared:  0.3634,    Adjusted R-squared:  0.3364
F-statistic: 13.42 on 2 and 47 DF,  p-value: 2.455e-05
```



The Pipe

$$X \rightarrow Z \rightarrow Y$$

Z is a "mediator"

X and Y are associated $\qquad Y \not\perp\!\!\!\perp X$

Influence of X on Y transmitted through Z

Once stratified by Z, no association $\qquad Y \perp\!\!\!\perp X \mid Z$

Example: Treatment → Fungus → Growth

$$X \rightarrow Z \rightarrow Y$$

```r
n <- 1000
X <- rbern( n , 0.5)
Z <- rbern( n , (1-X)*0.1 + X*0.9 )
Y <- rbern( n , (1-Z)*0.1 + Z*0.9 )
```

```
        Y
X       0   1
  0   430  87
  1    93 390
```

$Y \not\perp\!\!\!\perp X$

```
> cor(X,Y)
[1] 0.64
```

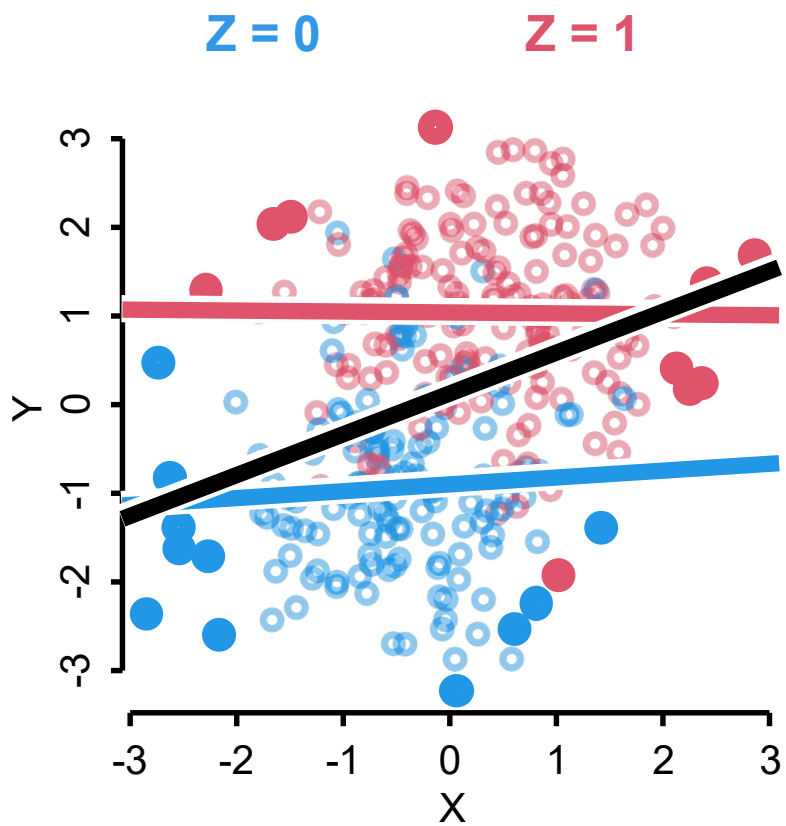
```
Z = 0
        Y
X       0   1
  0   422  39
  1    53   5

Z = 1
        Y
X       0   1
  0     8  48
  1    40 385
```

$Y \perp\!\!\!\perp X \mid Z$

```
> cor(X[Z==0],Y[Z==0])
[1] 0.002
> cor(X[Z==1],Y[Z==1])
[1] 0.052
```

$$X \rightarrow Z \rightarrow Y$$

```
cols <- c(4,2)

N <- 300
X <- rnorm(N)
Z <- rbern(N,inv_logit(X)) Y <-
rnorm(N,(2*Z-1))

plot( X , Y , col=cols[Z+1] , lwd=3 )

abline(lm(Y[Z==1]~X[Z==1]),col=2,lwd=3)

abline(lm(Y[Z==0]~X[Z==0]),col=4,lwd=3)

abline(lm(Y~X),lwd=3)
```



Pipe Example

Plant growth experiment

100 plants

Half treated with anti-fungal

Measure growth and fungus

Estimand: Causal effect of treatment on plant growth

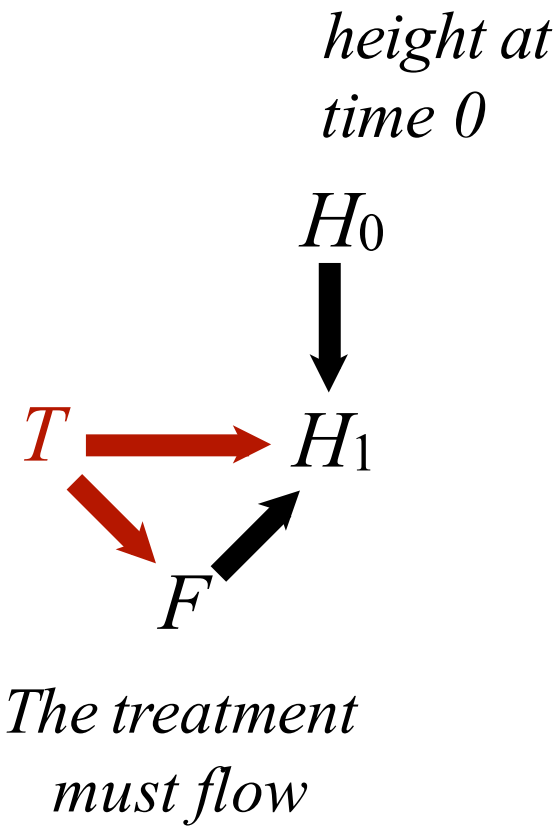
Statistical model

Estimand: Total causal effect of T

The path T –> F –> H_1 is a pipe

Should we stratify by F?

NO — that would block the pipe



The Collider

$$X \rightarrow Z \leftarrow Y$$

Z is a "collider"

X and *Y* are not associated (share no causes) $Y \perp\!\!\!\perp X$

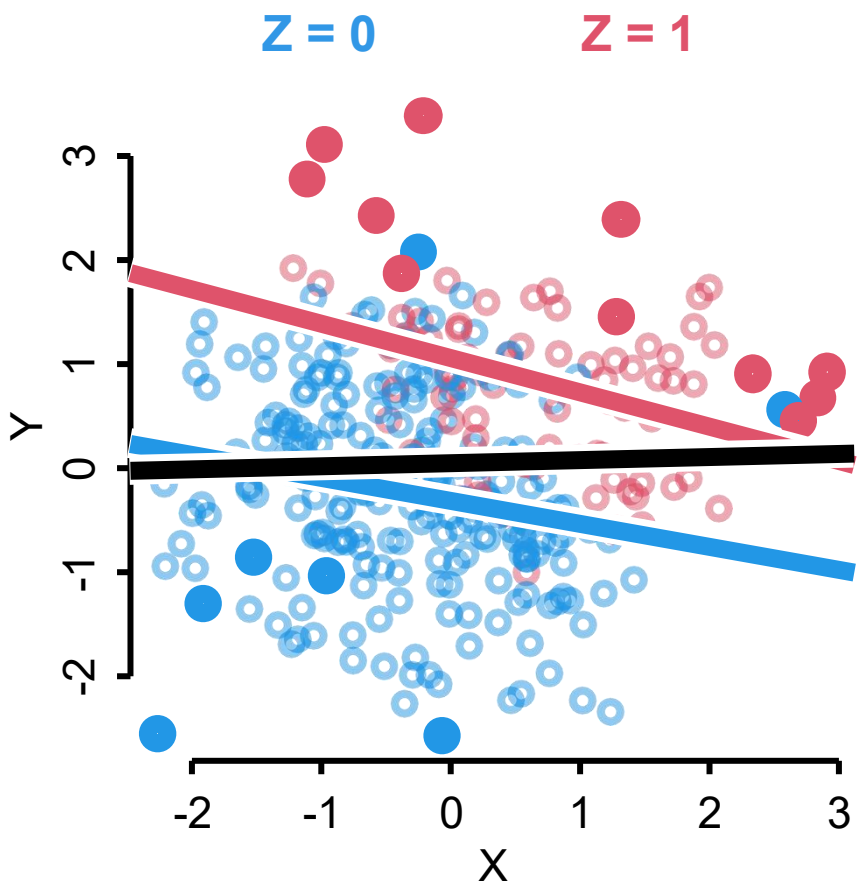
X and *Y* both influence *Z*

Once stratified by *Z*, *X* and *Y* associated $Y \not\perp\!\!\!\perp X \mid Z$

Example: Trustworthy→Selection←Newsworth

$$X \rightarrow Z \leftarrow Y$$

```
cols <- c(4,2)

N <- 300
X.  <- rnorm(N)
Y.  <- rnorm(N)
Z.  <-  rbern(N,inv_logit(2*X+2*Y-2))

plot( X , Y , col=cols[Z+1] , lwd=3 )

abline(lm(Y[Z==1]~X[Z==1]),col=2,lwd=3)

abline(lm(Y[Z==0]~X[Z==0]),col=4,lwd=3)

abline(lm(Y~X),lwd=3)
```



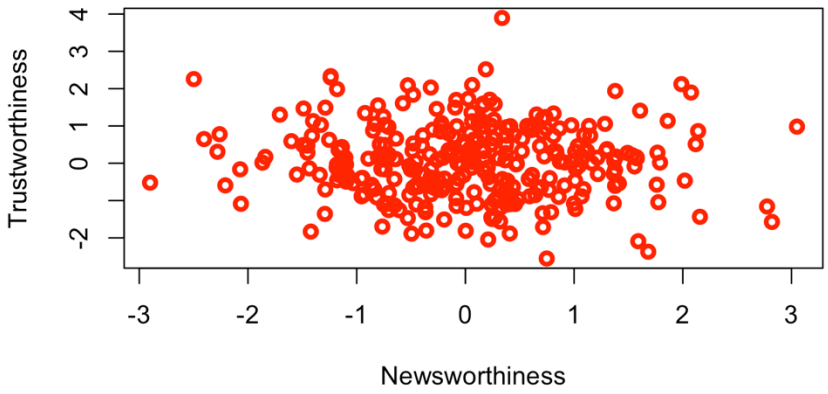
Collider example

Some biases arise from selection

Suppose: 200 grant applications

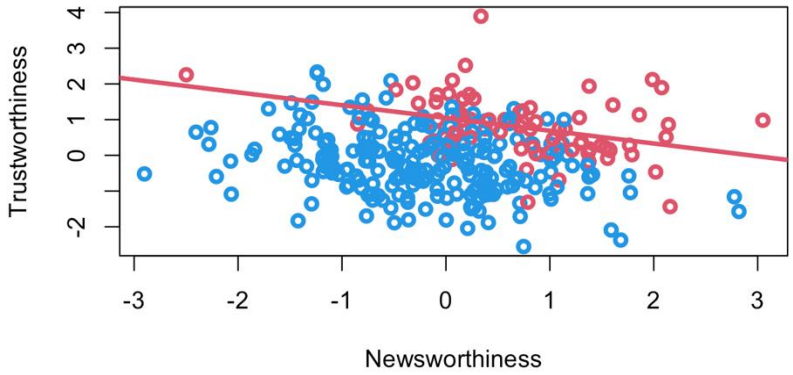
Each scored on newsworthiness and trustworthiness

No association in population



Collider example

- Some biases arise from selection
- Suppose: 200 grant applications
- Each scored on newsworthiness and trustworthiness
- No association in population
- Strong association after selection



Collider examples

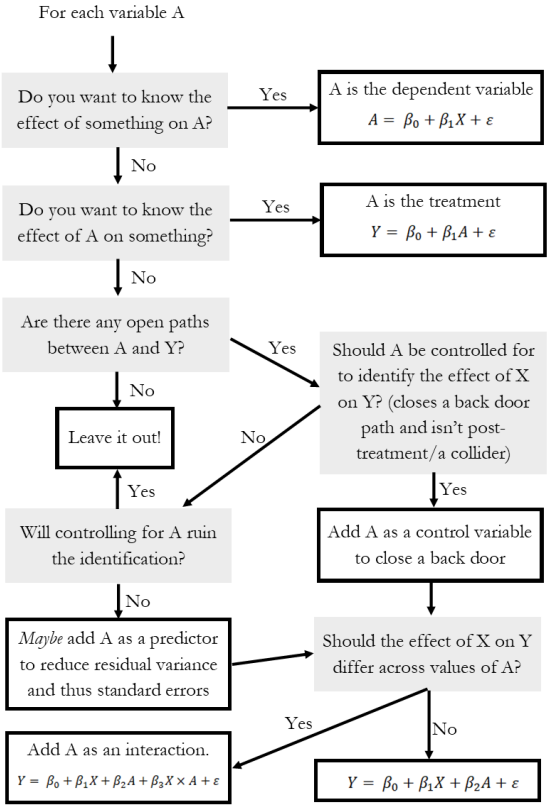
- Similar examples:

- Restaurants survive by having good food or a good location => bad food in good locations

- Actors can succeed by being attractive or by being skilled => attractive actors are less skilled

- Stratified by marital status, negative association between age and happiness

What should you do?

- On the diagram, we're thinking about what to do with a variable .
- Is it the outcome variable, or is some other variable the outcome?
- Is it the treatment or is some other variable than the treatment.
- If it's neither, where does it go on the diagram and in a regression, if anywhere at all?

- Source:
- https://theeffectbook.net/ch-StatisticalAdjustment.html